

① Fundamental concepts used in web development

① XML

XML is a markup language used to store, organize and exchange data b/w different applications. It allows developers to define their own tags.

ex:

```
<student>
  <name> sai </name>
  <course> cse </course>
</student>
```

② JSP

JSP is a server-side technology based on Java used to create dynamic web pages. JSP allows Java code and HTML to work together.

③ MVC Architecture

MVC stands for model-view controller. It divides an application into three parts.

- Model: handles data and business logic.
- View: displays information to the user.
- Controller: Receives user requests and co-ordinates the model and view.

Purpose: makes applications easier to develop, maintain and modify.

Q.1) XSLT

XSLT is used to transform XML documents into other formats, such as HTML, XML or text.

Q.2) XML schema

XML schema commonly called XSD, defines the structure and rules of an XML document. It specifies and attributes, data types and relationships.

Q.3) JSTL

JSTL provides predefined tags that simplify common tasks in JSP such as loops, conditions, formatting and database related operations.

Q.4) PHP

PHP is a server-side scripting language used to develop dynamic and interactive web applications.

Q.5) Database connectivity

Database connectivity allows a web application to communicate with a database to store, retrieve, update and delete information.

• JSPC for Java applications

• PHP/MySQL for PHP applications.

⑧ Parser

* Parser analyzes and processes structured data such as XML, HTML or programming language code according to its syntax rules.

② Inter connection of these technologies in a web application.

* These technologies work together to provide data storage, validation, processing, presentation and communication in a web application.

⑨ XML → XML schema

XML is used to represent structured data. XML schema (XSD) defines the rules that the XML document must follow. The XML document is validated against the schema to ensure that its data types are correct.

⑩ XML → XSLT

After XML data is processed, XSLT can transform the XML into HTML or another XML format so that the data can be displayed to users.

iii) MVC → Application organization.

MVC provides the overall structure of the application.

user → controller → Model → database/XML →

Model → view → user.

- The controller receives the user's request.
- the model performs business logic and handles data.
- the database connectivity layer communicates with the database.
- XML parsers can process XML data.

iv) database connectivity → data processing

Jsp/Java or PHP can use database connectivity mechanisms to communicate with databases. Data can be retrieved, processed, updated or stored based on user requests.

overall flow

user Request → controller → Jsp/PHP → business logic →

→ database/XML → parser → data processing →

XSLT/ Jsp view → HTML Response → user.